

Six-Week Internship Daily Work Log

Web Development and AI-Agent-Assisted Coding | 6 Weeks | 6 Working Days per Week

Week 1 - Web Development Foundations

Day	Today's Duties	Working On	Technology	Key Learning
Day 1	Reviewed internship goals and analyzed the structure of a modern website.	Planning the website layout and development workflow.	HTML5, CSS3, VS Code	Learned how requirements are converted into a clear page structure.
Day 2	Created semantic page sections including header, navigation, main content, and footer.	Building the basic website skeleton.	HTML5, Semantic HTML	Understood how semantic tags improve structure and accessibility.
Day 3	Styled page elements using typography, spacing, colors, borders, and backgrounds.	Designing a clean and consistent user interface.	CSS3, Box Model	Learned how the CSS box model controls layout and spacing.
Day 4	Built flexible rows, columns, navigation menus, and aligned content blocks.	Creating responsive page sections.	CSS Flexbox	Learned efficient alignment and distribution using Flexbox.
Day 5	Developed a responsive card layout that adapts to desktop and mobile screens.	Improving layout responsiveness.	CSS Grid, Media Queries	Understood breakpoints and mobile-first responsive design.
Day 6	Reviewed the first website version and corrected visual and structural issues.	Refining the static frontend.	Chrome DevTools, HTML, CSS	Learned to inspect elements and debug styling problems systematically.

Week 2 - JavaScript and Interactive Web Pages

Day	Today's Duties	Working On	Technology	Key Learning
Day 1	Revised JavaScript variables, functions, arrays, objects, and control flow.	Preparing logic for interactive features.	JavaScript ES6	Strengthened understanding of reusable functions and data structures.
Day 2	Selected page elements and changed text, styles, and classes through scripts.	Adding dynamic behavior to the webpage.	JavaScript DOM API	Learned how JavaScript reads and updates the Document Object Model.
Day 3	Implemented click, input, and submit event handlers for interface controls.	Making forms and buttons interactive.	JavaScript Events	Understood event-driven programming and user interaction handling.
Day 4	Added client-side validation for required fields, email format, and input length.	Building a reliable contact form.	JavaScript, HTML Forms	Learned to validate data and show clear error messages.
Day 5	Fetches sample data from an external endpoint and displays it as cards.	Connecting the frontend to online data.	Fetch API, JSON, REST API	Learned asynchronous requests and JSON response handling.
Day 6	Handled loading, success, empty, and error states for API-based content.	Improving the API user experience.	Async/Await, Try/Catch	Learned robust error handling for asynchronous operations.

Week 3 - React Frontend Development

Day	Today's Duties	Working On	Technology	Key Learning
Day 1	Set up a React project and studied components, JSX, props, and project flow.	Migrating the interface to React.	React, Vite, Node.js	Learned component-based development and reusable UI design.
Day 2	Converted repeated interface sections into reusable React components.	Creating headers, cards, buttons, and forms.	React Components, JSX	Understood how component reuse improves consistency and maintenance.
Day 3	Managed form values, toggles, and selected items using component state.	Adding stateful interactions.	React useState Hook	Learned how state updates trigger interface re-rendering.
Day 4	Loaded API data when components mounted and synchronized results with state.	Displaying dynamic data in React.	React useEffect, Fetch API	Learned how effects handle external data and lifecycle behavior.
Day 5	Created multiple pages and enabled navigation without full page reloads.	Building a single-page application.	React Router	Understood client-side routing and route-based component rendering.
Day 6	Tested responsive behavior and fixed component spacing and state issues.	Polishing the React frontend.	React DevTools, Chrome DevTools	Learned to debug component state and responsive UI defects.

Week 4 - Backend, APIs, and Data Handling

Day	Today's Duties	Working On	Technology	Key Learning
Day 1	Created a basic server and organized routes for application features.	Starting the backend service.	Node.js, Express.js	Learned how a server receives requests and sends responses.
Day 2	Built GET, POST, PUT, and DELETE endpoints for sample records.	Developing CRUD API operations.	Express.js, REST API	Understood HTTP methods and resource-based API design.
Day 3	Connected the frontend form and listing pages with backend endpoints.	Integrating frontend and backend.	Fetch API, Express.js, JSON	Learned end-to-end data flow between browser and server.
Day 4	Designed data fields and stored structured records in a database.	Creating persistent application data.	MySQL, SQL	Learned tables, primary keys, and basic data modeling.
Day 5	Added backend validation and handled invalid requests and server errors.	Improving API reliability.	Express Middleware, SQL	Learned server-side validation and consistent error responses.
Day 6	Tested API endpoints and corrected response, routing, and database issues.	Verifying backend functionality.	Postman, Node.js, MySQL	Learned systematic API testing and backend debugging.

Week 5 - Coding with AI Agents

Day	Today's Duties	Working On	Technology	Key Learning
Day 1	Studied how AI coding agents interpret goals, inspect code, and propose changes.	Understanding AI-assisted development workflows.	AI Coding Agents, Prompting	Learned that clear context and constraints improve generated solutions.
Day 2	Wrote structured prompts with goals, existing behavior, expected output, and limits.	Creating reliable implementation instructions.	Prompt Engineering, Markdown	Learned to reduce ambiguity by defining acceptance criteria.
Day 3	Used an AI agent to generate a feature plan before modifying the project.	Planning a new web feature safely.	AI Agent, Git, Project Documentation	Learned to review architecture and risks before approving code changes.
Day 4	Reviewed AI-generated code and corrected logic, naming, and integration problems.	Validating agent-created frontend code.	React, JavaScript, AI Agent	Learned that AI output requires human verification and testing.
Day 5	Used an AI agent to diagnose a bug from logs, code flow, and observed behavior.	Debugging an application issue.	AI Agent, Browser DevTools, Logs	Learned to provide evidence and reproduction steps for accurate diagnosis.
Day 6	Created checkpoints and compared code changes before accepting the final solution.	Controlling and reviewing agent modifications.	Git, Version Control, AI Agent	Learned safe rollback, change review, and incremental implementation.

Week 6 - AI-Assisted Project Completion

Day	Today's Duties	Working On	Technology	Key Learning
Day 1	Combined frontend, backend, and database features into one working application.	Completing full-stack integration.	React, Node.js, Express.js, MySQL	Learned how application layers coordinate through defined interfaces.
Day 2	Asked an AI agent to refactor repeated code and improve component organization.	Improving code quality and maintainability.	AI Agent, React, JavaScript	Learned to evaluate refactoring by behavior, readability, and test results.
Day 3	Added loading indicators, validation messages, and error feedback across the app.	Improving usability and reliability.	React, CSS, API Error Handling	Learned that clear system feedback creates a better user experience.
Day 4	Tested important user flows on desktop and mobile and fixed discovered defects.	Performing end-to-end quality checks.	Manual Testing, DevTools, Responsive Design	Learned to test complete workflows rather than isolated components.
Day 5	Prepared the project for deployment and checked environment and configuration values.	Creating a production-ready build.	Vite Build, Environment Variables, Git	Learned the difference between development and production configuration.
Day 6	Documented completed work, technologies used, challenges, and final outcomes.	Finalizing the internship project and report.	Markdown, Git, Project Documentation	Learned to communicate technical work clearly and record future improvements.